Azure Big Data Architecture Design | Healthcare

Selected scenario: Regional Health Network | 15 hospitals and 200+ clinics | 800 GB/day

1. Data Sources & Ingestion Layer

Regional Health Network operates 15 hospitals and more than 200 clinics and receives approximately 800 GB/day (24 TB/month) of electronic health records (EHR), imaging, laboratory results and device telemetry. Design capacity for 20% annual growth, while separating low-latency clinical events from bulk historical loads. Azure Data Factory orchestrates incremental Epic extracts using approved FHIR/HL7 interfaces, private connectivity and change-data capture where supported. Event Hubs ingests high-volume HL7/FHIR event feeds, while IoT Hub authenticates and manages connected devices. Imaging objects enter through secure batch transfer; metadata and references are cataloged without unnecessarily duplicating large images. Ingestion validates schemas, timestamps, patient identifiers and provenance, quarantines malformed records and records replay offsets. A queue-backed interface protects Epic from upstream processing spikes; no analytics pipeline writes directly to the clinical system of record.

2. Storage Layer

Azure Data Lake Storage Gen2 (hierarchical namespace) provides encrypted, access-controlled bronze/raw, silver/validated and gold/curated zones. Partition datasets by event date, facility and approved data domain; use Delta Lake tables for ACID updates, deduplication and auditable version history. A starting steady-state planning model stores roughly 24 TB hot (first 30 days), 72 TB cool (next 90 days) and 144 TB archive (older infrequently retrieved snapshots), subject to retention policy and actual data reduction. Only archive datasets that tolerate hours-long retrieval; active clinical and research datasets remain online. Lifecycle policies tier eligible objects and delete them only after legal, clinical and research holds expire. Encryption in transit and at rest, private endpoints, Microsoft Entra ID role-based access, managed identities, Azure Key Vault, Microsoft Purview classification and immutable audit retention support HIPAA safeguards. A signed business associate agreement, documented risk assessment and minimum-necessary access remain mandatory; Azure services alone do not establish compliance.

3. Processing Layer

Azure Databricks is the primary Spark/Delta processing engine because it unifies complex EHR normalization, streaming transformations and scalable feature engineering more flexibly than a warehouse-first Synapse-only design; HDInsight would impose greater cluster-management overhead. Structured Streaming consumes Event Hubs data, applies clinical rules and writes curated events. Independently deployed Azure Functions processes urgent notifications, keeping bedside alert delivery outside long-running batch jobs. Scheduled Databricks jobs reconcile encounters, lab values and imaging metadata; Azure Data Factory manages dependencies and quality gates. Begin with autoscaling job clusters of 4-16 workers, benchmark throughput against peak loads and enforce automatic termination. Isolate production streaming from research jobs and reserve baseline compute only after utilization stabilizes. Monitor freshness, dropped events, latency, error rates and cost with Azure Monitor and Log Analytics.

4. Analytics & Output Layer

Azure Machine Learning manages versioned population-risk and readmission models, monitored for drift, calibration, subgroup performance and data leakage. De-identified or approved limited datasets enable governed research workspaces, with review and export controls. Power BI provides facility-level dashboards for care gaps, admissions, length of stay and cohort trends; row-level security limits access. Approved APIs expose model scores to care-coordination applications and Epic workflows, without silently changing patient records. Rule-based critical thresholds trigger clinical notifications through a resilient, acknowledged delivery service with escalation, audit trails and human verification. Target p95 event-to-alert latency below five seconds for validated device feeds, 99.9% analytics-platform availability and explicit disaster-recovery objectives; these are proposed acceptance targets, not guarantees. Clinical safety testing and clinician sign-off precede deployment.

5. Data Flow Summary, Costs & Implementation

Epic, laboratory, imaging and device sources flow through Data Factory, Event Hubs and IoT Hub into the raw lake; Databricks validates and enriches events into curated Delta tables, while Azure ML models and Power BI or approved APIs consume only authorized outputs. Urgent alerts use an independent low-latency processing path. For a US Central,

pay-as-you-go planning model, assume 730 hours/month and the tier volumes above. Indicative monthly allowances are shown below; they are NOT a saved Azure Pricing Calculator result and must be recalculated using region-specific SKUs, VM/DBU hours, ingestion throughput, Power BI licenses and transaction counts before submission or procurement.

Estimated monthly cost component	USD/month
Storage: 24 TB hot + 72 TB cool + 144 TB archive	$1,656
Databricks compute + DBUs	$19,000
Event Hubs, IoT Hub and alert compute	$4,500
Data Factory orchestration	$2,700
Azure ML training / inference	$4,800
Power BI and serving	$2,500
Networking / transfer	$2,200
Security, Key Vault, governance / monitoring	$2,600
Backup and disaster recovery	$3,400
Estimated monthly total	**$43,356**

Use 30-day budget alerts, daily cost dashboards, cluster shutdown schedules, storage tiering, reserved VM capacity for predictable workloads and Databricks pre-purchase only when justified. With modeled 15% annual cost growth, three-year operating spend is approximately $1.81 million, excluding implementation staffing, licensing changes, taxes and disaster recovery exercises. Deliver a 90-day pilot at two hospitals, followed by security/Epic integration validation, clinical acceptance testing, phased rollout and staff training. Risks include poor identity matching, false alerts, data egress, governance gaps and cost overruns; mitigate with reconciliation, alert tuning, private networking, independent audits and staged deployment.

Pricing references: https://azure.microsoft.com/en-us/pricing/calculator/ | https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/pricing-calculator | https://azure.microsoft.com/en-us/pricing/details/databricks/